

Exp No: 5 ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS

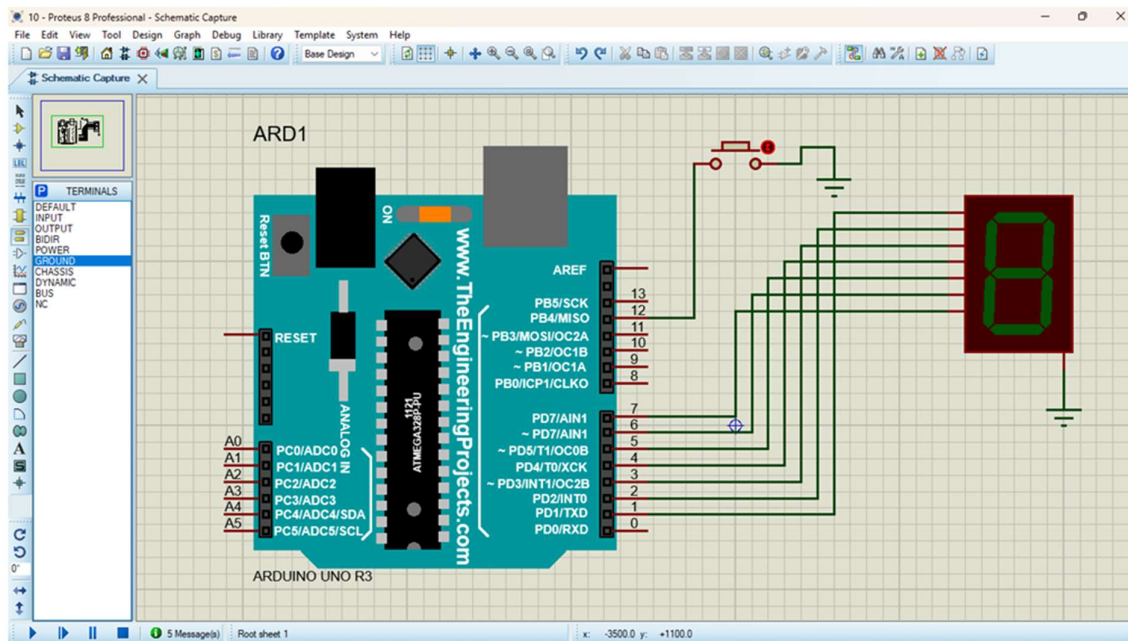
AIM:

To write an Embedded C program for Manual Electronic counter using Arduino IDE and Proteus

SOFTWARE REQUIRED:

- Proteus 8 software
- Arduino IDE

CIRCUIT DIAGRAM:



PROGRAM:

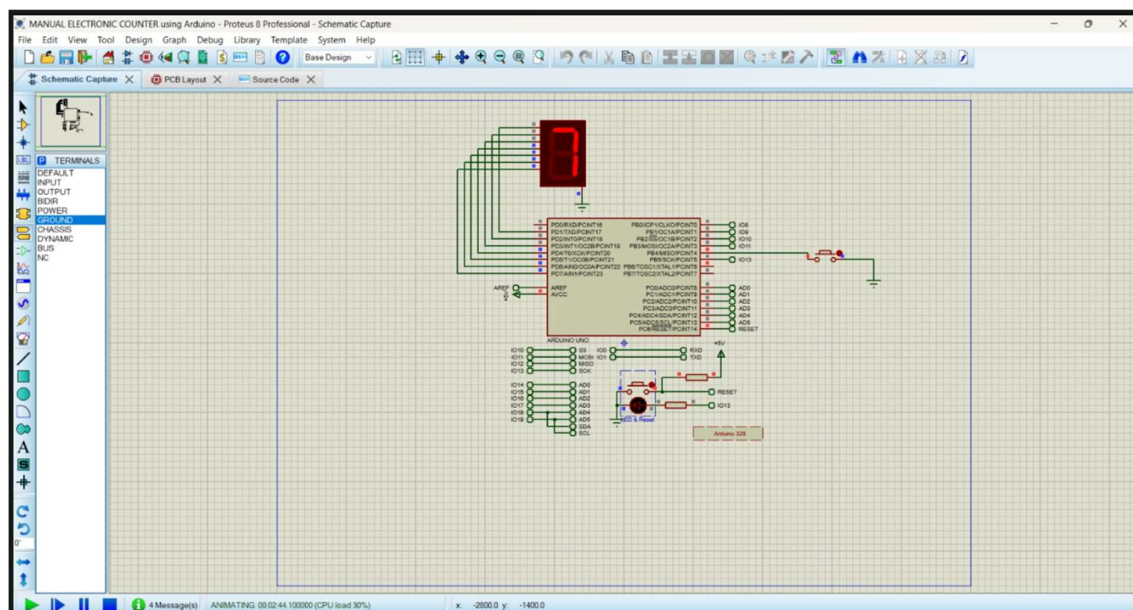
```

// Program Sketch
int trig = 10;
int echo = 9;
long duration;
int cm;
void setup() {
  // put your setup code here, to run once:
  pinMode(trig, OUTPUT);
  pinMode(echo, INPUT);

  Serial.begin(9600);
}
void loop() { // Sketch to calculate ultrasonic distance
  digitalWrite(trig, LOW);
  delayMicroseconds(10);
  digitalWrite(trig, HIGH);
  delayMicroseconds(10);
  digitalWrite(trig, LOW);
  delayMicroseconds(10);
  duration = pulseIn(echo, HIGH);
  cm = (duration/2) * 0.034;
  // Print Value on Serial Monitor
  Serial.print(" Distance = ");
  Serial.print(cm);
  Serial.println(" cm");
}

```

OUTPUT:



RESULT:

Thus the program has been successfully verified and executed.